A* Search Algorithm — Python Code

```
def astar(graph, start, goal, heuristic):
    frontier = [(heuristic[start], 0, start, [start])]
    visited = []

    while frontier:
        frontier.sort()
        f, cost, node, path = frontier.pop(0)

        if node in visited:
            continue

        visited.append(node)
        print("Visiting:", node, "g =", cost, "f =", f)

        if node == goal:
            return path, cost

        for child, edge_cost in graph[node]:
            if child not in visited:
                new_cost = cost + edge_cost
                new_f = new_cost + heuristic[child]
                new_path = path + [child]
                frontier.append((new_f, new_cost, child, new_path))

    return None, float('inf')


graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('D', 2), ('E', 5)],
    'C': [('F', 2), ('G', 5)],
    'D': [],
    'E': [('H', 1)],
    'F': [],
    'G': [('I', 1)],
    'H': [],
    'I': []
}

heuristic = {
    'A': 7,
    'B': 6,
    'C': 4,
    'D': 7,
    'E': 3,
    'F': 3,
    'G': 1,
    'H': 2,
    'I': 0
}

path, cost = astar(graph, 'A', 'I', heuristic)

if path:
    print("Path:", " -> ".join(path))
    print("Path cost:", cost)
else:
    print("No path exists")
```